in the boiling point is known as elevation of boiling point. A plot of vapour pressure versus temperature for water and an aqueous solution is given below

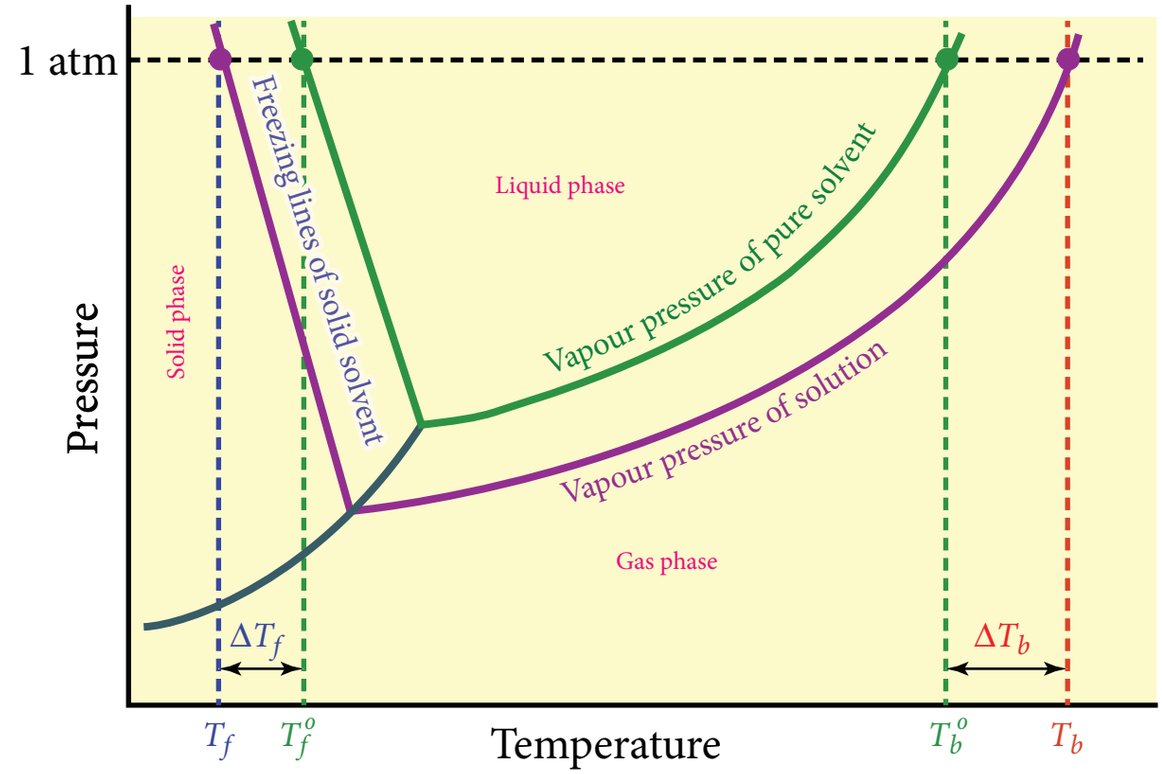


Figure 9.11 Elevation of boiling point and depression in freezing point

The vapour pressure of the solution increases with increase in temperature as shown in the above figure. The variation of vapour pressure with respect to temperature for pure water is given by the violet coloured curve. At 100 ^{0}C the vapour pressure of water is equal to 1 atm. Hence the boiling point of water is 100 ^{0}C ($T_b°$). When a solute is added to water, the vapour pressure of the resultant solution is lowered. The variation of vapour pressure with respect to temperature for the solution is given by green curve. From the graph, it is evident the vapour pressure of the solution is equal to 1 atm pressure at the temperature T_b which is greater than $T_b°$. The difference between these two temperatures (T_b-$T_b°$) gives the elevation of boiling point.

The elevation of boiling point (ΔT_b)= T_b - $T_b°$

The elevation of boiling point is directly proportional to the concentration of the solute particles.

$\Delta T_b \alpha m$ ------ (9.23)

m is the concentration of solution expressed in molality.

$\Delta T_b = K_b m$ ------ (9.24)

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